

Fleet Management App: Structure, Pages & Estimated Timeline

This document outlines the proposed structure, key pages/screens, and rough development time estimates for the Flutter fleet management application, incorporating distinct **Admin, Driver, and Customer** roles.

1. Overall App Structure:

The app will be designed to present different interfaces and functionalities based on the logged-in user's role (Admin, Driver, Customer).

- **Authentication Flow:** Handles user login (identifying the role), registration (potentially separate paths for different roles), and password recovery.
- **Role-Based Main Interface:** After login, the app will display a tailored interface:
 - **Admin/Manager Interface:** Likely uses a Bottom Navigation Bar or Drawer for access to comprehensive management features (Dashboard, Vehicles, Drivers, Customers, Map, Maintenance, Reports, Settings).
 - **Driver Interface:** Simplified interface, possibly using a Bottom Navigation Bar, focusing on assigned tasks (e.g., Current Task/Route, Start/End Report, Vehicle Info, Profile/Settings).
 - **Customer Interface:** Focused on booking, tracking their specific service/vehicle, viewing history, and managing their profile. Could use a Bottom Navigation Bar (e.g., Book, Track, History, Profile).
- **Modular Features:** Core data (Vehicles, Drivers, Users, Bookings) will be shared, but access and manipulation will be role-dependent.

2. Key Pages/Screens by Role:

- **Authentication (Common to all):**
 - **SplashScreen:** Checks authentication status, determines role.
 - **LoginPage:** Unified login, potentially directing to role-specific flows post-login.
 - **RegistrationPage:** May need role selection or separate registration paths.
 - **ForgotPasswordPage.**
- **Admin/Manager Role:**
 - **AdminDashboardPage:** High-level overview (vehicles, drivers, ongoing trips, revenue hints, alerts).

- VehicleListPage: Manage all vehicles.
- VehicleDetailsPage: View/edit vehicle details, assign drivers, view history.
- AddEditVehiclePage.
- DriverListPage: Manage all drivers.
- DriverDetailsPage: View/edit driver details, assign vehicles, view history/reports.
- AddEditDriverPage.
- CustomerListPage: Manage customer accounts.
- CustomerDetailsPage: View customer info and booking history.
- MapViewPage: Overview map with filtering options for all active vehicles/drivers/routes.
- MaintenanceLogPage: Manage maintenance for all vehicles.
- AddEditMaintenancePage.
- ReportsPage: Access comprehensive reports (fleet performance, driver logs, financials, etc.).
- AdminSettingsPage: App-wide settings, user role management.

- **Driver Role:**

- DriverDashboardPage: Shows current assignment (vehicle, route/task), key stats, next steps.
- StartTripReportPage: Form to start a trip/shift, requiring **odometer reading** and vehicle check.
- EndTripReportPage: Form to end a trip/shift, requiring **final odometer reading**, trip notes.
- CurrentTaskPage: Details of the active route or task, potentially with navigation integration.
- TripHistoryPage (Driver): View own past trips and reports.
- VehicleInfoPage (Driver): View details of the currently assigned vehicle.

- DriverProfilePage: View/edit own profile information.
- NotificationsPage (Driver): Receive assignments, alerts.

- **Customer Role:**

- CustomerDashboardPage / BookRidePage: Interface to request/book services or view ongoing bookings.
- TrackBookingPage: Real-time map view of the assigned vehicle for their current booking. ○ BookingHistoryPage: List of past services/bookings.
- CustomerProfilePage: Manage own profile, payment methods (if applicable).
- SupportPage: Contact support or provide feedback.

- **Common (Conceptual):**

- MainLayoutWrapper: Adapts navigation (Bottom Bar/Drawer) based on the logged-in role.
- LogoutFunctionality.

3. Core Features & Functionality

Customer (Employee) Core Features

Roster Management:

- Create and manage daily/weekly transportation schedules
- Specify pickup and drop-off locations with timing preferences
- Set recurring schedules for regular commutes
- Real-time availability updates and schedule modifications
- Automatic roster optimization based on office hours

Trip Management & Tracking:

- View upcoming scheduled trips with driver assignments
- Real-time trip tracking with live driver location
- Driver information display including photo, rating, and contact details
- Trip status notifications (started, delayed, en route, completed)
- Trip cancellation and rescheduling based on availability

Dashboard & Analytics:

- Personal transportation statistics and usage patterns
- Visual representation of trip history and performance metrics
- Cost analysis and transportation efficiency metrics

- Monthly and yearly travel summaries

Safety & Communication:

- SOS emergency button for immediate assistance
- Real-time notifications for trip updates and alerts
- Direct communication channel with assigned driver
- Emergency contact integration and automated alerts

Driver Core Features

Shift & Vehicle Management:

- Automated shift reminders 30 minutes before login time
- Mandatory photo capture for customer identification and verification
- Vehicle assignment confirmation and condition reporting
- Digital shift start/end reporting with comprehensive logs

Trip Operations:

- Route optimization for customer groups (3-5 employees per trip)
- Traffic-aware routing with real-time adjustments
- Automated customer notifications for trip status updates
- ETA calculations and arrival time communications

Customer Service:

- Real-time location sharing with assigned customers
- Trip progress updates and delay notifications
- Group pickup coordination and efficient routing
- Customer feedback and rating system integration

Reporting & Documentation:

- Trip completion reports with detailed logs
- Vehicle condition and maintenance alerts
- Daily activity summaries and performance metrics
- Incident reporting and emergency response protocols

Admin Core Features

Fleet Operations Management:

- Vehicle inventory with real-time status tracking
- Driver assignment with automated scheduling
- Live fleet monitoring dashboard with GPS tracking
- Maintenance scheduling and service alerts
- Emergency response coordination and incident management
- Route planning and optimization oversight

Driver Management & Oversight:

- Driver profile creation and document verification
- Performance monitoring with ratings and feedback analysis
- Shift scheduling and assignment management
- Driver behavior tracking (speed, route adherence, feedback)
- Training records and certification tracking
- Performance improvement and disciplinary action tracking

Customer & Employee Management:

- Employee account creation with role-based access
- Service request handling and resolution tracking
- Roster approval and modification oversight
- Trip booking management and conflict resolution
- Employee feedback monitoring and satisfaction analysis
- Corporate client account and billing management

Analytics & Business Intelligence:

- Operational dashboards with real-time KPIs
- Fleet utilization and cost efficiency reports
- Driver performance evaluation with scorecards
- Customer satisfaction monitoring and trend analysis
- Revenue tracking and profitability analysis
- Fuel consumption and environmental impact reporting
- Peak usage analysis and capacity planning

System Administration & Control:

- User role management and permission assignment
- Route approval with traffic pattern analysis
- Notification system and automated alert configuration
- Corporate HR and payroll system integration
- Compliance monitoring and regulatory reporting
- System backup and data security management

Financial Management:

- Cost tracking per trip, vehicle, and driver
- Budget management and expense monitoring
- Invoice generation and billing automation
- Fuel cost analysis and optimization
- Insurance claim and accident cost tracking
- Revenue forecasting and financial planning

Emergency & Safety Management:

- Real-time SOS alert monitoring and response
- Incident reporting with investigation tracking

- Safety protocol enforcement and compliance
- Emergency contact management and alerts
- Vehicle breakdown coordination and rescue operations
- Safety training schedule and compliance tracking

5. Rough Development Timeline

Development Progress & Timeline

✓ COMPLETED TASKS:

- **Project Setup & Basic Structure: 3-5 days ✓**
 - Application architecture and framework setup
 - Database design and initial configuration
- **Authentication Flow (Handling Roles): 5-8 days ✓**
 - Multi-role login system implementation
 - User registration and password recovery
 - Role-based access control setup
- **Admin Module - Basic UI Framework: 2-3 days ✓**
 - Admin dashboard basic UI structure
 - Navigation and layout components
 - Role-based UI rendering

CURRENTLY IN PROGRESS:

- **Customer Module - Basic UI Development: 2-3 days 🔄 IN PROGRESS**
 - Customer dashboard basic UI structure
 - Navigation and booking interface components
 - Profile management UI screens

UPCOMING DEVELOPMENT TASKS:

Admin Module Features:

- **Admin - Vehicle Management: 7-12 days**
 - Vehicle inventory and registration system
 - Vehicle assignment and tracking
- **Admin - Driver Management: 6-10 days**
 - Driver profile and assignment management
 - Performance monitoring and reporting
- **Admin - Customer Management: 4-6 days**
 - Employee account management
 - Service history and profile oversight
- **Admin - Map & Basic Tracking: 7-10 days**
 - Real-time vehicle tracking implementation
 - Route visualization and monitoring
- **Admin - Maintenance Module: 5-8 days**

- Maintenance scheduling and record keeping
- Vehicle condition monitoring
- **Admin - Basic Reporting: 5-7 days**
 - Performance analytics and dashboard creation
 - Basic reporting functionality

Driver Module Features:

- **Driver - Dashboard & Trip Reporting (incl. Odometer): 6-10 days**
 - Driver dashboard with assignment overview
 - Trip start/end reporting with vehicle checks
 - 30-minute login reminders and photo capture
- **Driver - Task/History/Profile: 4-6 days**
 - Route assignment and navigation (3-5 customer groups)
 - Trip history and performance tracking
 - Driver profile management

Customer Module Features:

- **Customer - Booking & Tracking: 8-14 days**
 - Roster creation with login/logout times and locations
 - Real-time trip tracking and driver information
 - Driver photo, contact details, and ratings display
- **Customer - History & Profile: 4-6 days**
 - Trip history and service records
 - Personal profile and preference settings
 - Visual statistics and analytics dashboard

Advanced Features:

- **Settings & Profiles (All Roles): 4-6 days**
 - Cross-platform profile management
 - System-wide settings and preferences
- **Real-time Updates & Notifications (Role-aware): 6-10 days**
 - Push notification system implementation
 - SOS emergency system integration
 - Real-time data synchronization
- **UI Polishing & Basic Testing: Ongoing, add ~20-30% to total time**
 - User interface refinement and optimization
 - Comprehensive testing across all user roles

Phase 5: Advanced Features & Integration (Weeks 21-24)

Settings & Profiles (All Roles): 4-6 days

- Cross-platform profile management
- System-wide settings and preferences

Real-time Updates & Notifications: 6-10 days

- Push notification system implementation
- Real-time data synchronization

UI Polishing & Testing: Ongoing throughout development

- User interface refinement and optimization
- Comprehensive testing and quality assurance

6. Current Development Status & Progress

✓ COMPLETED TASKS:

- Role-based authentication system with multi-user login
- User registration and password recovery functionality
- Role-based access control implementation
- Application architecture and database setup
- Admin module basic UI framework and navigation
- Customer module basic UI structure (currently in development)

CURRENTLY WORKING ON:

- Customer module UI components
- Customer dashboard interface development
- Booking and roster management UI screens

UPCOMING TASKS:

- Complete customer module functionality
- Admin module feature implementation
- Driver module development
- Real-time tracking and notifications

7. Estimated Timeline Summary

Estimated Total (Core MVP with 3 Roles): ~ 20 - 28 Weeks

Current Progress Summary:

- **Completed:** Foundation setup and role-based authentication + Admin UI framework (~2.5 weeks)
- **In Progress:** Customer module UI development (~2-3 days remaining)
- **Remaining:** ~17.5-25.5 weeks for full MVP completion

Important Considerations:

- Managing state and permissions across three distinct roles adds significant complexity
- Customer-facing roster management and tracking system requires extensive user experience design
- Real-time features and notification systems require robust backend infrastructure
- Driver photo verification and group route optimization (3-5 customers) are critical features

- SOS emergency system integration is essential for safety compliance
- Comprehensive testing across all user roles is essential for successful deployment

“I am planning to complete it within the specified days as per the schedule. If I complete it earlier, I will inform you immediately. However, since this involves several coding tasks, there is a possibility it may take a few additional days. I will keep you updated on the progress accordingly.”